

One direct result of using prisms is that we have a linear projection that maps the **France** embedding vector to capital city candidates’ unembedding vectors. We think this linear projection is meaningful not just for **France**, but has similar effects on other country tokens too. To check this hypothesis, we apply the same projection matrix to other countries’ embedding vectors. Figure 5 shows the same matrix does indeed project other country names to their respective capitals.

This suggests that the network learns to represent country names and capital city names in such a way that it can easily transform a country embedding to the capital city unembedding using a linear projection. We hypothesize that this observation can be generalized to other relations encoded by the network as well.

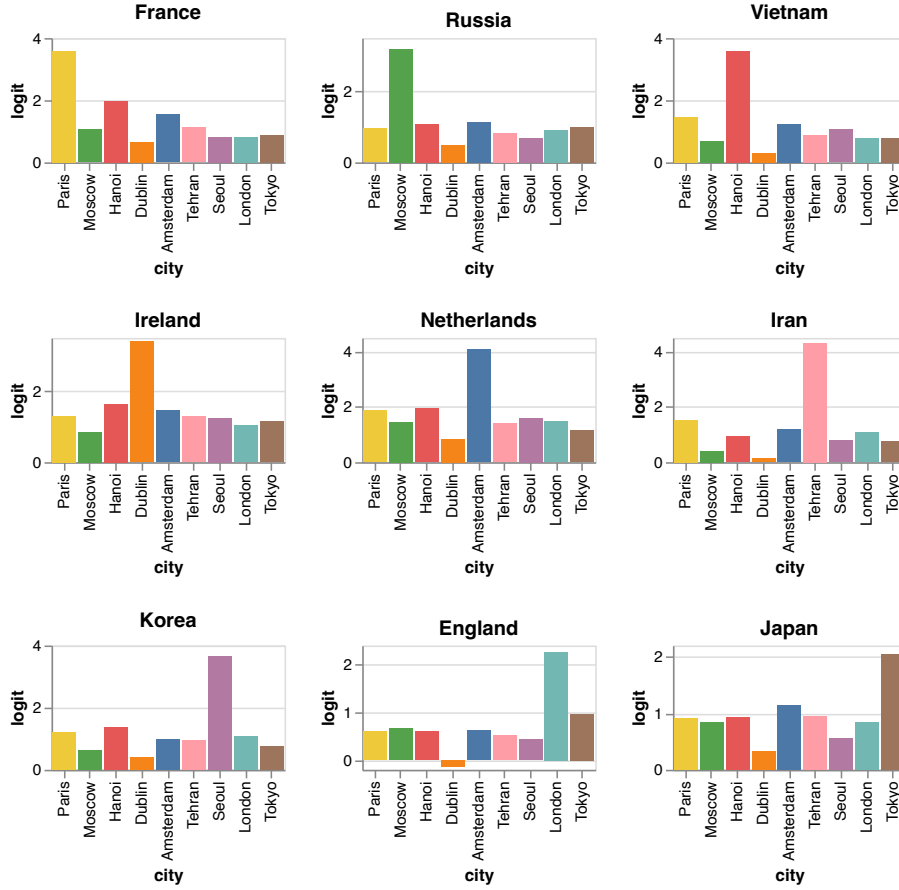


Figure 5: Linear projection from country embedding to capital city’s logit.

Digit addition. Let’s explore how **gemma-2b** performs arithmetic by asking it

to complete the following:

7+2=_

The model correctly predicts 9 as the next token. To understand how it achieves this, we employ our prisms toolbox. First, using the residual prism, we decompose the residual stream and examine the contributions of different layers for target tokens ranging from 0 to 9 (Figure 6). The MLP layer at layer 16 (m16) stands out, predicting 9 with a significantly higher logit value than other candidates. This substantial gap is unique to m16, indicating its crucial role in the model’s prediction of 9.

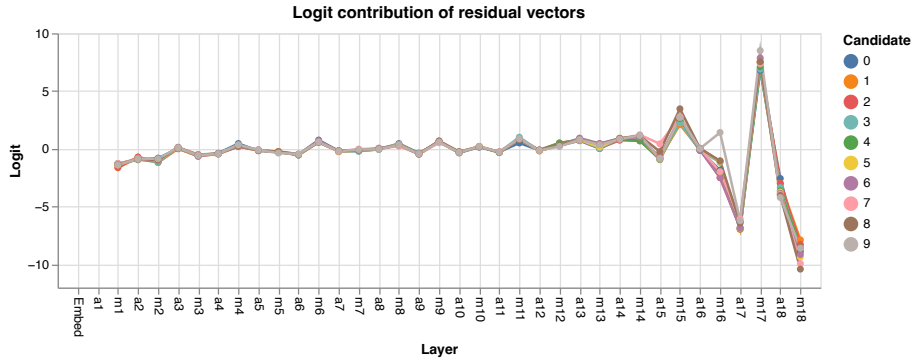


Figure 6: Contributions of different layers to the logit outputs of different candidates (from 0 to 9) using the residual prism.

Next, we use the MLP prism to identify which neurons in m16 drive this behavior. Decomposing m16 into contributions from its 16,384 neurons, we find that most are inactive. Extracting the top active neurons, we observe that they account for the majority of m16’s activity. Figure 7 shows these top neurons’ contributions to candidates from 0 to 9, revealing distinct patterns for each neuron. For example, neuron 10029 selectively differentiates odd and even numbers. Neuron 11042 selectively predicts 7, while neuron 12552 selectively avoids predicting 7. Neurons 15156 and 2363 show sine-wave patterns. While no single neuron dominantly predicts 9, the combined effect of these neurons’ predictions peaks at 9.

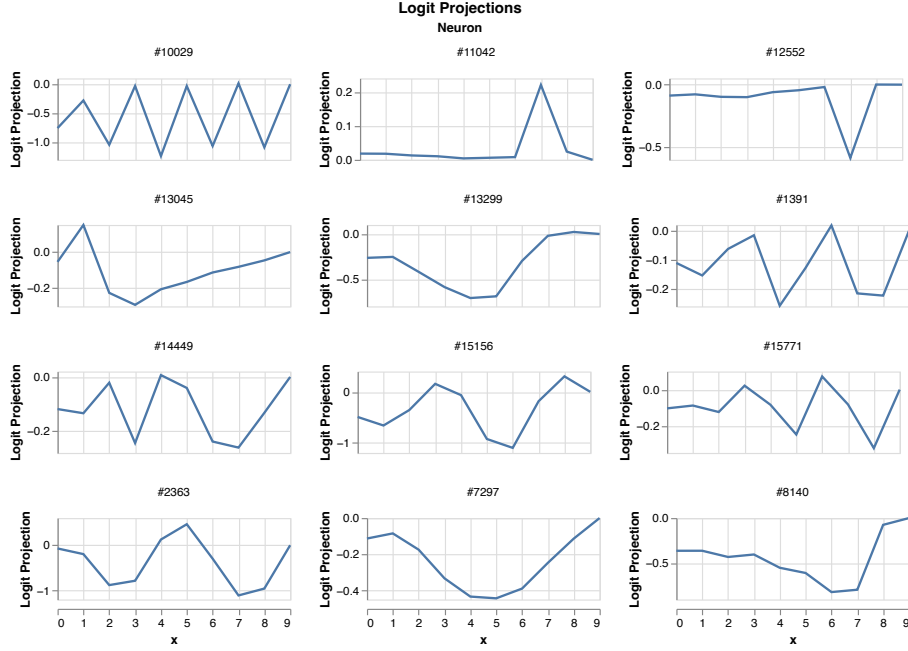


Figure 7: Top neuron contributions for different targets ranging from 0 to 9.

Note that the neurons' contributions to the target logits are simply linear projections onto different target token unembedding vectors. The neuron activity patterns in Figure 7 are likely encoded in the target token unembeddings; as such, these patterns can be easily extracted using a linear projection. When we visualize the digit unembedding space in 2D (Figure 8), we discover that the numbers form a heart-like shape with reflectional symmetry around the 0-5 axis.